

JOINS AND SET OPERATIONS

```
MySQL 8.0 Command Line Cli  ×  +  ▾

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.22 MySQL Community Server - GPL

Copyright (c) 2000, 2020, Oracle and/or its affiliates. All rights reserved.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE joins_sets_db;
Query OK, 1 row affected (0.03 sec)

mysql> USE joins_sets_db;
Database changed
mysql> CREATE TABLE students (
  ->     student_id INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     department_id INT,
  ->     marks INT
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO students VALUES
  -> (1, 'Rahul', 1, 85),
  -> (2, 'Anita', 2, 78),
  -> (3, 'Kiran', 1, 92),
  -> (4, 'Priya', 3, 88),
  -> (5, 'Arjun', 2, 75);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```



```
mysql> CREATE TABLE departments (  
->     department_id INT PRIMARY KEY,  
->     department_name VARCHAR(50),  
->     hod VARCHAR(50)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO departments VALUES  
-> (1, 'CSE', 'Dr. Rao'),  
-> (2, 'ECE', 'Dr. Kumar'),  
-> (3, 'IT', 'Dr. Sharma'),  
-> (4, 'AIML', 'Dr. Reddy');
```

Query OK, 4 rows affected (0.02 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM students;
```

student_id	name	department_id	marks
1	Rahul	1	85
2	Anita	2	78
3	Kiran	1	92
4	Priya	3	88
5	Arjun	2	75

5 rows in set (0.00 sec)



```
mysql> SELECT * FROM departments;
```

department_id	department_name	hod
1	CSE	Dr. Rao
2	ECE	Dr. Kumar
3	IT	Dr. Sharma
4	AIML	Dr. Reddy

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT students.student_id,  
->      students.name,  
->      departments.department_name,  
->      departments.hod,  
->      students.marks  
-> FROM students  
-> INNER JOIN departments  
-> ON students.department_id = departments.department_id;
```

student_id	name	department_name	hod	marks
1	Rahul	CSE	Dr. Rao	85
2	Anita	ECE	Dr. Kumar	78
3	Kiran	CSE	Dr. Rao	92
4	Priya	IT	Dr. Sharma	88
5	Arjun	ECE	Dr. Kumar	75

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT students.student_id,  
->         students.name,  
->         departments.department_name,  
->         departments.hod,  
->         students.marks  
-> FROM students  
-> LEFT JOIN departments  
-> ON students.department_id = departments.department_id;
```

student_id	name	department_name	hod	marks
1	Rahul	CSE	Dr. Rao	85
2	Anita	ECE	Dr. Kumar	78
3	Kiran	CSE	Dr. Rao	92
4	Priya	IT	Dr. Sharma	88
5	Arjun	ECE	Dr. Kumar	75

5 rows in set (0.00 sec)

```
mysql> SELECT students.student_id,  
->         students.name,  
->         departments.department_name,  
->         departments.hod,  
->         students.marks  
-> FROM students  
-> RIGHT JOIN departments  
-> ON students.department_id = departments.department_id;
```

student_id	name	department_name	hod	marks
1	Rahul	CSE	Dr. Rao	85
3	Kiran	CSE	Dr. Rao	92
2	Anita	ECE	Dr. Kumar	78
5	Arjun	ECE	Dr. Kumar	75
4	Priya	IT	Dr. Sharma	88
NULL	NULL	AIML	Dr. Reddy	NULL

6 rows in set (0.00 sec)

```
mysql> SELECT students.name,  
->         departments.department_name  
-> FROM students  
-> CROSS JOIN departments;
```

name	department_name
Rahul	CSE
Rahul	ECE
Rahul	IT
Rahul	AIML
Anita	CSE
Anita	ECE
Anita	IT
Anita	AIML
Kiran	CSE
Kiran	ECE
Kiran	IT
Kiran	AIML
Priya	CSE
Priya	ECE
Priya	IT
Priya	AIML
Arjun	CSE
Arjun	ECE
Arjun	IT
Arjun	AIML

20 rows in set (0.00 sec)



```
mysql> SELECT
->     s1.name AS student1,
->     s2.name AS student2,
->     s1.department_id
-> FROM students s1
-> JOIN students s2
-> ON s1.department_id = s2.department_id
-> AND s1.student_id < s2.student_id;
```

```
+-----+-----+-----+
| student1 | student2 | department_id |
+-----+-----+-----+
| Rahul    | Kiran    | 1             |
| Anita    | Arjun    | 2             |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT name
-> FROM students
-> WHERE department_id = 1
->
-> UNION
->
-> SELECT name
-> FROM students
-> WHERE department_id = 2;
```

```
+-----+
| name  |
+-----+
| Rahul |
| Kiran |
| Anita |
| Arjun |
+-----+
```

```
4 rows in set (0.00 sec)
```

```

MySQL 8.0 Command Line Cli
mysql> SELECT name
-> FROM students
-> WHERE marks > 80
-> INTERSECT
-> SELECT name
-> FROM students
-> WHERE department_id = 1;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'INTERSECT'

SELECT name
FROM students
WHERE department_id = 1' at line 5
mysql> SELECT DISTINCT s1.name
-> FROM students s1
-> INNER JOIN students s2
-> ON s1.name = s2.name
-> WHERE s1.marks > 80
-> AND s2.department_id = 1;
+-----+
| name |
+-----+
| Rahul |
| Kiran |
+-----+
2 rows in set (0.01 sec)

```

```

mysql> SELECT name
mysql> SELECT name
-> FROM students s1
-> WHERE marks > 80
-> AND NOT EXISTS (
->     SELECT 1
->     FROM students s2
->     WHERE s2.name = s1.name
->     AND s2.department_id = 1
-> );
+-----+
| name |
+-----+
| Priya |
+-----+
1 row in set (0.00 sec)

mysql> |

```